

# PROJECT SYNOPSIS

---

## Title of the Project

---

VDCET Official Website with AI-Powered RAG Knowledge Assistant

---

### Student & Team Information

- **Lead Student:** Suraj Gajanan Pathade

- **Team Members:**

1. Suraj Gajanan Pathade
2. Rohit Shirbhate
3. Khushal Hajare
4. Harshal Kalbhut
5. Sanket Nasare
6. Tejas Sawargaokar

- **Course:** B.Tech (Computer Engineering)

- **College:** Vilasrao Deshmukh College of Engineering & Technology, Mouda (VDCET)

- **Affiliated University:** Dr. Babasaheb Ambedkar Technological University (DBATU), Lonere

- **Guide:** Rohini Ma'am

- **Academic Session:** 2026–2027

---

## 1. Introduction

---

Vilasrao Deshmukh College of Engineering & Technology (VDCET), Mouda (DTE Code: 04141) requires a modern, responsive, and easy-to-navigate official digital portal. This project delivers an Apple-style web platform integrated with an AI-Powered Retrieval-Augmented Generation (RAG) Assistant and a hidden Content Management System (CMS).

---

## 2. Problem Statement

---

Traditional college websites are often overcrowded, difficult to navigate, unoptimized for mobile devices, and lack real-time query resolution. Standard AI chatbots frequently generate hallucinated or unverified facts (such as fake fees, dates, or faculty names). There is a need for a modern website equipped with a grounded AI

assistant that answers strictly from an official VDCET knowledge base.

---

### 3. Objectives

---

1. Build a minimal, Apple-style responsive website for VDCET Mouda.
  2. Develop a RAG AI assistant grounded in verified VDCET knowledge chunks.
  3. Provide source citations for AI answers and return strict fallback notifications when queries exceed knowledge boundaries.
  4. Implement a hidden, secure admin portal (/admin) for simple notice, event, and document management.
  5. Donate the completed, production-ready website to VDCET as a B.Tech Computer Engineering project gift.
- 

### 4. Methodology & Architecture

---

The system is built using Next.js 14 (App Router), TypeScript, and Tailwind CSS. The backend utilizes a modular repository layer (vdcet-db.json / SQLite) storing notices, events, documents, and knowledge base chunks.

When a user submits a query on /ai-assistant:

1. The RAG engine calculates relevance scores against knowledge chunks.
2. If relevance score passes threshold, top chunks are injected into a grounded LLM system prompt (Groq Llama-3 API).
3. The AI returns a grounded answer accompanied by official source badges.
4. If no chunk matches, the AI outputs: *"I couldn't find this information in the official VDCET knowledge base. Please contact the college office."*

Administrative management is isolated at secret route /admin protected by HTTP-only JWT sessions.

---

### 5. Technology Stack

---

- **Frontend:** Next.js 14, TypeScript, Tailwind CSS, Lucide React
  - **Backend & API:** Next.js API Routes, Server Actions
  - **Database:** Modular Repository Pattern (`vdcet-db.json` / SQLite)
  - **Security:** JWT Session Cookies (`jsonwebtoken`), Password Verification
  - **AI RAG Framework:** Groq SDK (`groq-sdk` Llama-3-8B-Instant) + TF-IDF Relevance Engine
-

## 6. Expected Outcome

---

A production-ready, high-performance website and AI assistant providing instant, accurate college information to prospective students and visitors while allowing non-technical college administrators to post notices easily.

---

## 7. Future Scope

---

- Voice search capabilities in English, Marathi, and Hindi.
  - Automated document OCR ingestion into the AI knowledge base.
- 

## 8. References

---

1. Next.js Documentation: <https://nextjs.org/docs>
2. State Common Entrance Test Cell, Maharashtra: <https://cetcell.mahacet.org/>
3. Dr. Babasaheb Ambedkar Technological University: <https://dbatu.ac.in/>